Cambodia Academy of Digital Technology
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G11-G1
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Course: Object-Oriented Programming (OOP)(Java)
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Project : Online Food Ordering System

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1. Introduction

1.1 Background of the Project

In today's digital era, technology is widely used to improve efficiency in daily activities. One common area is food ordering in restaurants. Many small restaurants still rely on manual systems such as writing orders on paper or taking verbal orders. These methods are slow and often lead to mistakes such as incorrect orders, missing items, and calculation errors.

To solve these problems, this project introduces an Online Food Ordering System developed using Java. The system simulates a digital ordering process where customers can select food items, add them to a cart, and place orders efficiently.

1.2 Objectives of the Project

The main objectives of this project are:

- To design and develop a food ordering system using Java
- To apply Object-Oriented Programming (OOP) concepts in a real application
- To allow customers to select and manage food orders easily
- To calculate total cost automatically with discount support
- To manage orders in a structured and organized way

1.3 Project Impact

This system helps improve accuracy and efficiency in food ordering. It reduces human errors in calculation and order handling. It also demonstrates how software solutions can replace manual processes in real-world scenarios.

2. Literature Review

2.1 Similar Systems

Many modern applications such as Foodpanda and GrabFood provide online food ordering services. These systems allow users to browse menus, place orders, and make payments digitally.

2.2 Inspiration and Extension

This project is inspired by real-world food ordering platforms but simplified for learning purposes. Unlike commercial systems, this project focuses on:

- Core ordering functionality
- Application of OOP principles
- System design and structure

It does not include advanced features such as online payment or delivery tracking, but it builds a strong foundation for future extension.

3. Implementation

3.1 Task Breakdown and Responsibilities

The project was developed as a team effort, with each member contributing to specific parts of the system.

- **Ngin Sinbe** focused on improving system functionality and applying OOP concepts such as method overriding. The `showInfo()` method in the Customer class was overridden to display user-specific information. In addition, discount calculation was implemented using method overloading in `FoodItem.setPrice()`, allowing flexible price updates with discounts.
- **Chea Meiya** worked on console output and system display. This includes formatting the menu, system messages, and receipt output to make them clear and readable. Improvements were also made to handle null values when searching for items, ensuring the system runs without errors.
- **Sou Pin** was responsible for system design and the implementation of core Object-Oriented Programming concepts. This includes structuring the project into different packages and defining relationships between classes such as User, Customer, and Admin. Polymorphism was implemented through inheritance (Customer extends User) and interface implementation (Order implements Payable), allowing flexible handling of different user types and payment calculations.
- **Lenh Sokmean** handled the cart and order processing logic. This includes implementing the functionality for adding items to the cart and creating orders. A key contribution is the order snapshot feature in the Order class, which stores item details at the time of purchase. This ensures that previous orders remain unchanged even if prices are updated later.

3.2 System Design

The system is developed using Java and follows Object-Oriented Programming principles. It is divided into three main packages:

- **model** → contains data classes (User, Customer, FoodItem, etc.)
- **logic** → handles system operations (Cart, Order, Menu)

- **service** → contains interfaces (Payable)

3.3 Project Structure and Components

Main Classes

- **User (Abstract Class)**
  Base class for all users in the system.
- **Customer (extends User)**
  Handles customer actions such as placing orders and viewing receipts.
- **Admin (extends User)**
  Manages system operations like updating food prices.
- **FoodItem**
  Stores food details such as name, price, and category.
- **Menu**
  Manages the list of available food items.
- **Cart**
  Stores selected food items and their quantities.
- **Order**
  Processes orders, calculates total cost, and prints receipts.
- **MembershipType (Enum)**
  Defines discount levels (NONE, SILVER, GOLD).
- **Payable (Interface)**
  Defines the method calculateTotal() for payment calculation.

3.4 Key Features

- Display food menu
- Add items to cart
- Update item quantities
- Calculate total price with discount
- Place and confirm orders
- Generate formatted receipt
- Admin can update food prices

3.5 OOP Concepts Applied

- **Encapsulation**
  Data and methods are contained within classes.
- **Inheritance**
  Customer and Admin inherit from User.

- **Polymorphism**
  Method overriding in showInfo() and interface implementation in Payable.
- **Abstraction**
  Abstract class User hides implementation details.

4. Results

4.1 Project Results

The system successfully implements all planned features:

- Menu display works correctly
- Cart operations function properly
- Orders are processed with correct calculations
- Discounts are applied based on membership type
- Receipts are generated with detailed information

4.2 Special Achievement

One important feature is the **Order Snapshot**:

- When an order is placed, item prices are saved
- Future price changes do not affect past orders

This ensures data consistency and reflects real-world system behavior.

4.3 Challenges Faced

- Managing cart data structure efficiently
- Ensuring correct price calculation after discount
- Handling null values when searching items
- Maintaining clean separation between classes

4.4 Lessons Learned

- Importance of proper class design
- How OOP improves code structure and maintainability
- Real-world application of programming concepts

5. Conclusion

5.1 Perspective

This project provided a strong understanding of Object-Oriented Programming concepts. It demonstrates how a real-world problem can be solved using structured programming techniques.

5.2 Challenges

The main challenges were designing class relationships and ensuring accurate data handling. These were solved through testing and improving code structure.

5.3 Final Thoughts

The Online Food Ordering System is a successful implementation of OOP concepts in Java. It provides a clear simulation of a real-world system and builds a strong foundation for future improvements such as database integration or graphical user interface.

6. Appendix

6.1 GitHub Repository

https://github.com/Soupin1205/OOP_Project.git

6.2 Diagrams

- Class Diagram (UML)
- System Flow Diagram